YEAR II SEMESTER 2 INDUSTRIAL TRAINING REPORT

PROPOSED COMMERCIAL BUILDING TO BE BUILT ON PLOT NO: 106 AT TEMEKE, DAR ES SALAAM

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DAR ES SALAAM *NOVEMBER, 2024.*

DECLARATION

I, SAILO, DICKSON E. hereby declare that the contents of this report are the results of my own efforts for attending industrial training and participating in various activities taking place at the site. I confirm that this work has not been presented elsewhere for a similar or any other degree award in any institution of higher learning.

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ACKNOWLEDGEMENT

I am overwhelmed and grateful to acknowledge the Almighty God for the gift of life, and for the wisdom and perseverance that he has bestowed upon me towards the beginning until the completion of this work despite of all the challenges, difficulties and discouragement that I have encountered in the whole process of its preparation.

I would like to express my sincere gratitude and appreciation to Ardhi University (ARU), and in a special manner The 8 Construction Limited for the assistance, cooperation and everyday support from the start to the completion of the whole period of my Industrial Training (IT).

I express my gratitude to my site supervisors; Qs. Anna Beatus, Forman Rama and the entire working team of technicians at the site for the close assistance and cooperation during the entire period of my training. Their technical assistance and teachings enabled me to acquire skills and knowledge which can be applicable in performing various construction activities.

In particular, I kindly pass my gratitude to our supervisor Ms. Mtitu from Ardhi University (ARU) department of Building Economics (BE) who came to the site for the supervision of our industrial training.

Lastly, I would like to address my special gratitude to my parents for their financial, moral and material support together with my fellow trainees who I cooperated and worked with during my industrial training for their participation in preparing this report.

ABSTRACT

This report was prepared to present the coverage of what was performed within eight weeks of industrial training conducted at Temeke site. The project was conducted at Serengeti, Temeke in Dar Es Salaam under The 8 Construction Limited supervision which involved the construction of eight storey building.

This report contains several chapters which shows the background of The 8 Construction Limited, its organization structure, objectives of the industrial training together with methodology and a summary of the activities conducted at the site.

The training focused on learning different construction activities pertaining field work such as; block work, concrete casting, formwork, steel fixing, plumbing, service installation, plastering and supervision of various site works. There was also participation in various activities as directed by the site supervisor.

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LIST OF ABBREVIATIONS

ARU – Ardhi University

PVC – Polyvinyl Chloride

IT – Industrial Training

BE – Building Economics

CHAPTER ONE

1.1 BACKGROUND OF THE 8 CONSTRUCTION LIMITED

The 8 Construction Limited is a Tanzanian Incorporated company and locally owned. The company offers engineering construction and construction project management. The company is dedicated to organizations or individuals seeking construction services.

The 8 construction company work closely with their clients to accurately interpret their dreams or visions in drawings and bring them to the desired reality through construction solutions to meet their needs. The company communicate with their clients throughout the construction process to keep them informed of progress and to ensure that the project keeps on schedule and within budget.

The company is committed to provide the best possible construction expertise and service to ensure cost effective and successful projects. Their in-house resources and capabilities allow them to provide complete and custom solutions. Furthermore, their strategic partnerships with vendors and suppliers allows them to successfully manage the unique needs of any project.

Activities Performed by The 8 Construction Limited

- Construction of new public buildings.
- Maintenance of public or private buildings.
- Allocation of houses to public servants.
- Selling of company houses to public servants on loan basis.
- Provision of building consultancy services to the public on all matters concerning building works.
- Leasing government houses on commercial basis.
- Provision of project management services for building projects.

1.2 VISION, MISSION AND INNOVATES

- **Vision**

  "To be the leading brand in construction recognized for trust, quality and innovation, supporting our staff and the communities in which we work, by being the contractor of choice in our market.''

- **Mission**

  "To be the contractor of choice by integrating the entire building lifecycle into a seamless platform to redefine how the world builds."

- **Innovates**

  "We anticipate challenges and proactively develop new capabilities to address the evolving needs of our clients. Our relentless focus on innovation leads to smarter building practices and technology that reduces complexity throughout our projects."

1.3 TANZANIA BUILDING AGENCY ORGANIZATION STRUCTURE

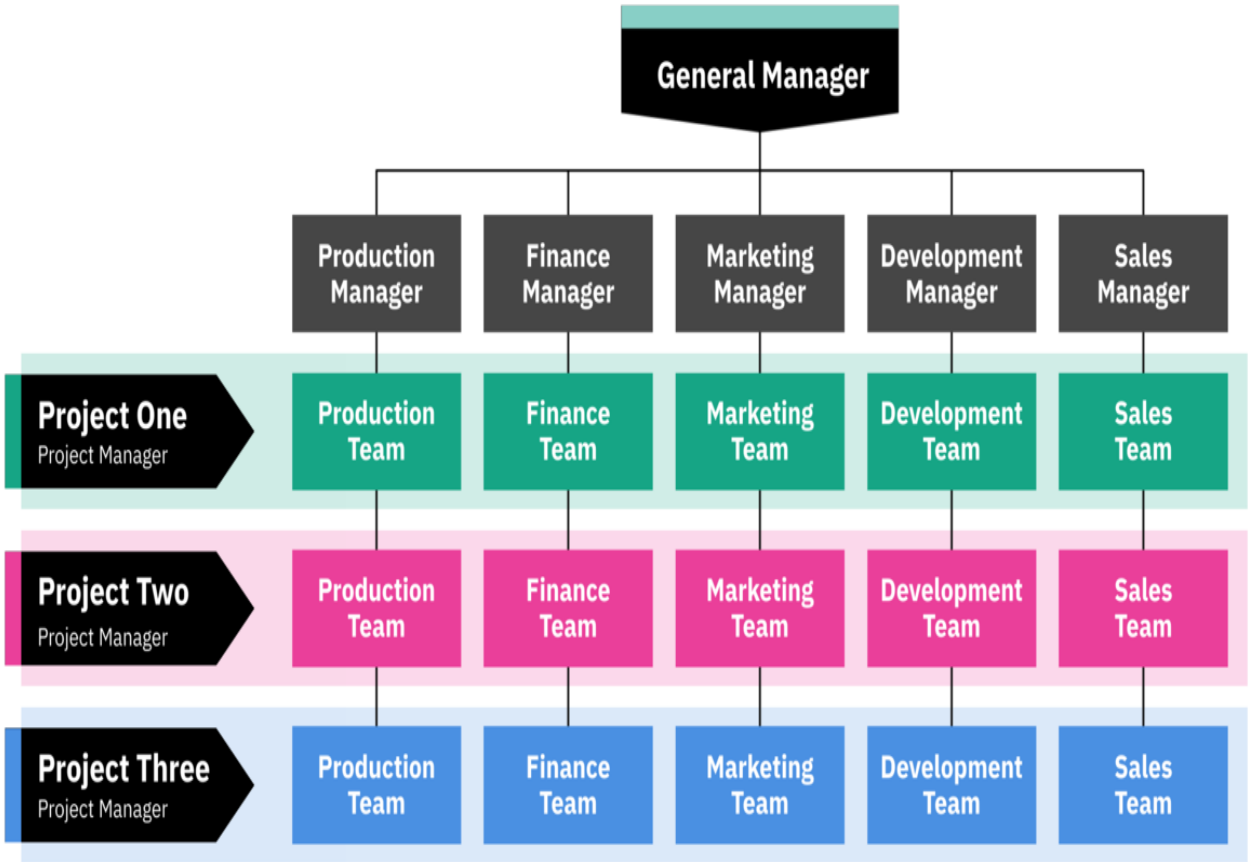


Figure 1: Showing Tanzania Building Agency organization structure

1.4 HEALTH AND SAFETY

Health and safety of workers at the site during the construction process was considered, this was due to the presence of personal protective gears such as hard hats and reflective jackets. There were also protective measures against trespassers using hoardings, warning tapes and guards. On top of these facilities, the safety officers were present at the site and also a readily available first aid kit was present at site. Several meetings were conducted for making awareness to all workers at the site concerning health and safety issues like wearing reflectors, safety shoes, helmets and safety belts.

1.5 OBJECTIVES OF INDUSTRIAL TRAINING

i. The industrial training period was aimed at giving us more knowledge about the construction industry.
ii. To transfer the knowledge obtained from the class to the site and be exposed to the career undertaken.
iii. To attain the experience of working as a group and improve communication skills among the co-members.
iv. To understand the importance of cost effective and the awareness of social and global environment as a quantity surveyor so as to make accurate predictions and assumptions.

1.6 METHODOLOGIES

The methodology involves carrying out various activities taking place in order to obtain data. During the industrial training course, I engaged in the following methods to collect the required data.

Physical observation, this involved taking pictures, asking questions and observing how the site engineer and technicians conducted their work according to the structural designs.

Total participation, this involved working together with the technicians and conducting other assigned activities by the site engineer such as steel fixing of reinforcement bars for slabs and columns, walling activities, curing and other assigned activities.

Consulting the engineers and other civil workers, this involved questioning the site engineer and the civil workers for more clarification on the activity taking place at the site.

CHAPTER TWO

2.1 EXISTING SITUATION AT THE SITE

2.1.1 Stage of the building

There was ongoing construction on the first floor of the building which included block work on the fence, steel fixing for slab, beam and staircase, and also formwork shuttering for the slab and staircase.

2.1.2 Temporary store room

The site had two store rooms, whereby one room was used as storage for construction tools such as; hammer, grinder and other safety tools, while the other room was used for the storage of building materials such as; marine boards, iron sheets cement, nails etc.

2.1.3 Temporary fence

There was also presence of a temporary fence made of corrugated iron sheets that surrounded all over the site.

2.1.4 Temporary guard house and gate

The site had one guard hut and one gate that was used as entrance for all the site workers.

CHAPTER THREE

SUMMARY OF ACTIVITIES PERFORMED AT THE SITE

3.1 Site Reconnaissance

This day we visited the site that we were about to conduct our industrial training, whereby the construction company The 8 Construction Limited had its project at Temeke. The project was a commercial building built at Temeke in Dar Es Salaam which involved the construction of a multi-storey building with eight (8) floors. At the site we found that they were done with the substructure part.



Plate 3.1: Showing the situation at the site



Plate 3.2: Showing the sign board at Temeke Project site

3.2 Drawing Interpretations

We got a chance of passing through different Architectural, Structural and Mechanical drawings such as; floor plans, foundation plans, door and window schedules, sections, top and bottom reinforcement plans, and other relevant drawings.

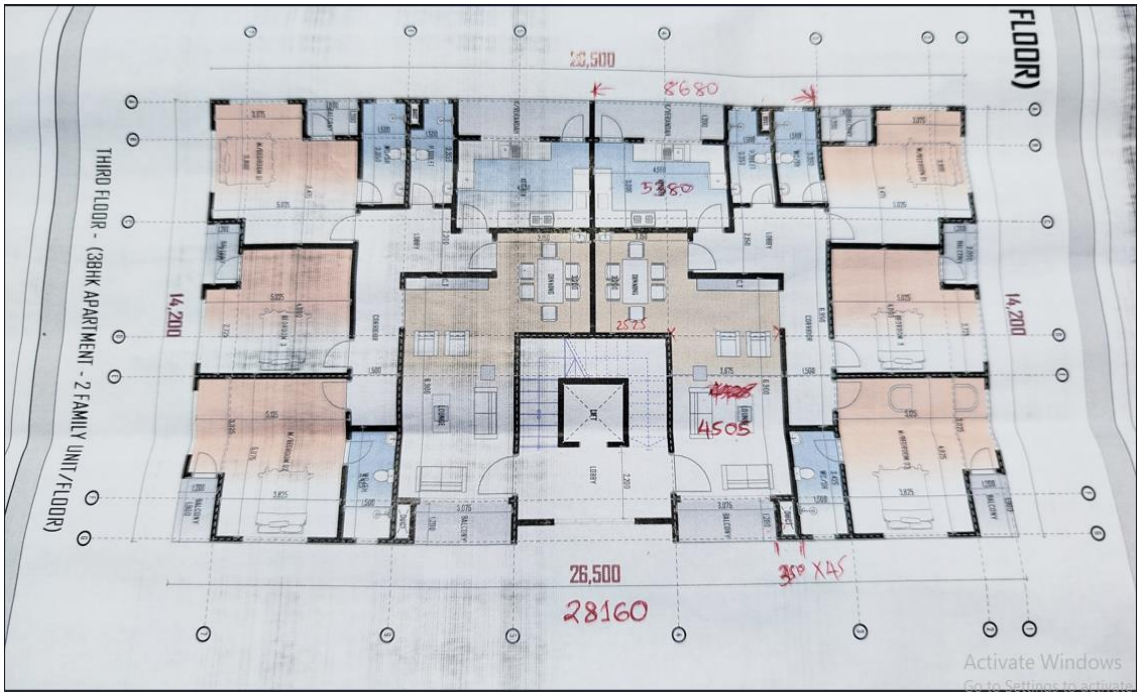


Plate 3.3: Showing a floor plan drawing

3.3 Formwork Shuttering

3.3.1 Preparing and fixing of formwork for the column

Formwork is the term given to a temporary mold into which concrete is placed to harden. Formwork for column consists of a vertical mound of a desired shape and size which has to retain the concrete and resist its initial pressure. Formwork can be of timber or metal material. At the site the formwork used was of timber molded to the dimension (3000 × 3000) mm which involved assembling of small sections of plywood joined by nails and supported by timbers (mirunda) on the outside.



Plate 3.4: Showing timber formwork for column

3.3.2 Preparing and fixing formwork for beams and slab

This involve materials such as plywood having (2400 × 1200) mm, crane rails as bridges held in position by timber struts, and marine board purposely fixed for maintaining the shape of the poured slab which were supported by vertically placed structures named props to support them during concrete casting and avoid bending of the slab members.



Plate 3.5: Showing a fixed beam formwork



Plate 3.6: Showing a fixed slab formwork

3.3.3 Preparing and fixing formwork for staircase

This part of the building is mainly located or situated for the purpose of linking one floor to the other. The type of formwork used at the site was timber formwork to construct concrete staircase, which were wood and marine boards. The stairs constructed were having tread of 300mm and riser of 150mm.



Plate 3.7: Showing fixed formwork for staircase

3.3.4 Preparing and fixing formwork for lift shaft

Lifts or elevators are usually enclosed by a shaft as it rises vertically through the building providing access from floor to floor apart from the staircase. The type of formwork used at the site for the construction of the lift shaft was timber formwork, which were wood and marine boards. The lift shaft is approximated to have (8 × 9) m respectively. The lift shaft contains two ducts, i.e. air condition and fire fighting in case of fire. The walls used for the lift shaft are shear, whereby the exterior shear wall was about 300mm and the partition wall having about 230mm.



Plate 3.8: Showing a fixed timber formwork for lift shaft

3.4 Steel Fixing

3.4.1 Preparing and fixing steel bars for columns

Tools used during the preparation include; gauge for bending rings, bending bar, binding wire and pincer. The activity carried out was the preparation of the reinforcement bars, whereby the column steel bars were 16mm thick, the ring space was 200mm and the bars used were 8mm thick for ring setting, followed by placing of the bars to their specific points.



Plate 3.9: Showing the fixing of reinforced bars for column

3.4.2 Preparing and fixing steel bars for beams and slab

The tools used to perform this activity were; pincer, bar bending, binding wire, tape measure, twisters and tongs. The steel reinforced bars used were of 16mm diameter which intersected each other and were secured or tied together using binding wire at a regular interval to create a stable slab and beam. It was then followed by fixing of space rebars and spacer blocks in order to maintain the required spacing between steel bars and the formwork, and ensure proper concrete cover and structural integrity.

During the fixation of reinforced steel bars for slab, a breathing was provided so as to provide room for internal forces within the structure to act without damaging the building.

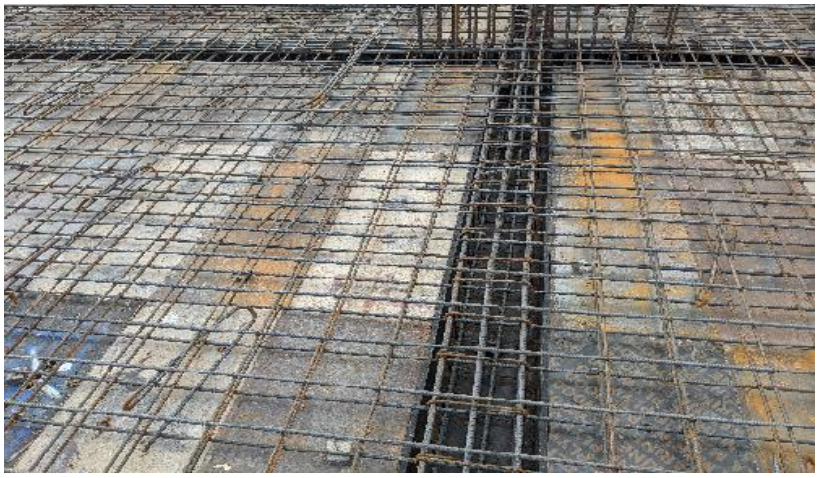


Plate 3.10: Showing fixed reinforced bars for beams and slab

3.4.3 Preparing and fixing steel bars for staircase

This part of the building is mainly located or situated for the purpose of linking one floor to the other. The activity carried out was the preparation of the reinforcement bars, whereby the staircase steel bars were 16mm thick. The stairs constructed were having tread of 300mm, riser of 150mm and an intersecting waist of 200mm.



Plate 3.11: Showing fixed reinforcement bars for stairs

- Reinforced bars play a big role in carrying the weight of all members i.e. the slab, beams, columns and shear walls. Therefore, for better cost minimization in the project it is advisable to reduce the size of the steel bars as ascending to the upper floors due to the decrease in the load.

3.5 Service Installation

This activity involved the installation of both electric conduit pipes for electrical services together with water and waste pipes from the wet zones i.e. kitchen and washrooms. The laying of pipes at the site was based on the concealed conduit wiring system, whereby there was neither risk of mechanical tear and wears nor fire damage in the cable insulation to be taken. The type of material used was Polyvinyl Chloride (PVC) pipes of 25mm diameter for data cables and fire alarm system, and that of 20mm diameter for electric cables. The use of PVC conduit pipes was the best choice because they don’t support corrosion.

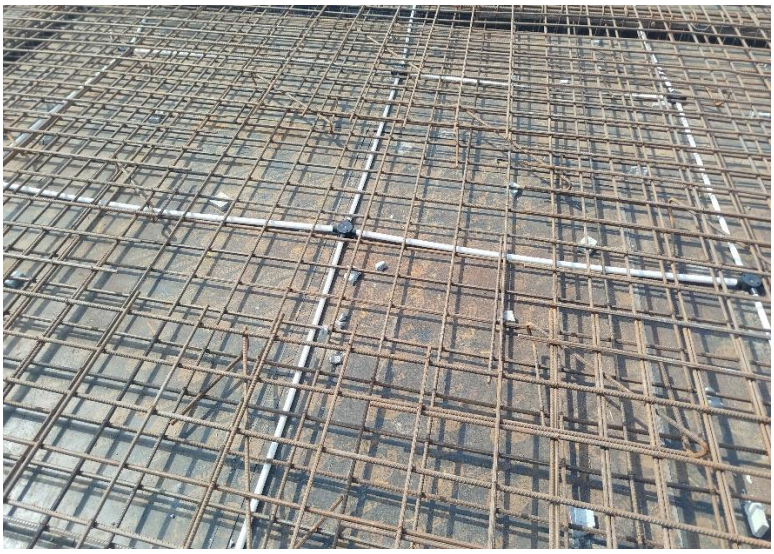


Plate 3.12: Showing fixed pipes for service installation

3.6 Concrete Casting

This process involved tools such as; concrete mixer truck, poker vibrator, wooden floats and spades. The concrete used for staircase, beams and slab was of grade 25 having the ratio 1:1.5:3 which was of 150mm thick while the one used for column and lift shaft was of grade 30. Concrete casted to beams, slab and staircase was pre cast, which was transported by the concrete mixer truck and connected to a compressor machine that pumped the mixture with a high pressure through a pipe to the specific floor concrete was required.

After casting the concrete, compaction process began which involved the use of a poker machine, as it's mostly used in large structures. The machine was submerged into the mixture of the casted concrete so as to remove air molecules and harden the compaction. It was then followed by the use of wooden float for further compaction and leveling.

Concrete was casted to the beams to attain the width of 230mm and depth of 230mm, and the slab having the thickness of 150mm. While the concrete casted to the columns had the dimensions of (150 × 300) mm and that for the shear walls had (150 × 600) mm.



Plate 3.13: Showing concrete casting



Plate 3.14: Showing leveling of concrete

3.7 Production of Concrete

The production of concrete at the site (insitu) was the one casted to columns and lift shaft which produced grade 30 with the ratio of 1:1:2 (cement, sand and aggregate).

Soon after the production, the concrete mixture which consists of cement, fine aggregates (sand), course aggregates and water at a required ratio and specific grade was then carried in buckets to the required area.



Plate 3.15: Showing a concrete mixer truck for insitu concrete production



Plate 3.16: Showing a concrete mixer truck carrying precast concrete

3.8 Curing of concrete

This refers to the method by which concrete is protected against loss of moisture required for hydration and kept within a recommended temperature range in order to increase the strength and decrease the permeability of the hardened concrete. Curing should be taken for a reasonable period of time in order for the concrete to achieve its potential strength and durability.

Curing process depends on several properties required for the concrete to be used for its particular purpose. And also, the ambient condition such as temperature and relative humidity of the surrounding atmosphere.

3.8.1 Objectives of concrete curing

- To maintain the required temperature within and around the concrete during a particular period of time.
- To maintain the water cement ratio required to hydrate all particles of cement and occupy the spaces in the pores.
- To provide enough quantity of water for the cement to hydrate and set completely.
- To increase the strength and durability of the casted concrete which helps to reduce the formation of cracks.

3.8.2 Methods of concrete curing

There are different methods of curing depending on the site constraints, the type of structure and other material parameters. These methods of water curing include sprinkling, ponding, wet covering and soaking.

i. Sprinkling method

This method is done continuously on the concrete surface and provides an efficient curing. It is mostly used in curing floor slabs. Concrete should be allowed to set sufficiently before sprinkling begins. Sprinkling can be done by the use of water pipe, especially if the construction is on a wide area such as a tall building, unlike on small constructions like residential houses sprinkling may be done by the use of hand and a bucket of water.

Vertical and sloping surfaces may be kept continuously wet by sprinkling water at their top surfaces and allowing it to run down the concrete structure.



Plate 3.17: Showing curing by sprinkling method

ii. Wet covering method

This is a widely used method of curing, particularly for structural concrete. Whereby, the exposed surface of concrete is prevented from drying out by covering it with gunny bags. The covering over vertical and sloping surfaces should be secured properly and also periodically wetted. The interval of wetting will depend upon the rate of evaporation of water.

It should be ensured that the surface of concrete is not allowed to dry even for a short time during the curing period. Most of the surfaces cured using this method were columns and shear walls. This method is very useful on wide construction projects as it saves economical cost compared to sprinkling method.



Plate 3.18: Showing curing by wet covering

iii. Soaking method

Soaking method involves curing of concrete by completely immersing it in water, typically in a tank or a pond. This method also ensures uniform hydration and often used for pre-cast concrete elements i.e. the spacer blocks and even while conducting concrete cube test.



Plate 3.19: Showing curing by soaking method

The period covered during curing normally ranges from 7 to 21 days depending on the weather and the type of concrete structure cured. Due to constant experiencing sunny condition at the site it took about 7 days for the column and shear walls to be cured and about 14 to 21 days for the slab and beams due to the variation in weather condition.

3.9 Walling

Wall is a vertical continuous structure that demarcates an area of land and bears the load which provides security and shelter to an enclosed area. The type of wall constructed at the site is masonry wall, whose construction was performed by using blocks of both 5 inches and 6 inches. The 6 inches' block was used only during the construction of the substructure while the 5 inches' block was used in the construction of the superstructure.

At the site, masonry work aimed in the construction of two types of wall i.e. partition walls and the load bearing walls, whereby the load bearing walls named shear walls supported the weight of the building while the partition walls or the non-load bearing walls divided the interior space into several rooms.

3.9.1 Objectives of walls

- To support upper floors and roofs.
- To provide thermal insulation to the building.
- To resist damp penetration into the building.
- To provide sufficient sound insulation.
- To prevent easily spread of fire.
- To provide privacy and security to the users.
- To divide and enclose the space.



Plate 3.20: Showing block work for partition wall

3.10 Bonding

This refers to the systematic way of arranging bricks or blocks in a brickwork or block work such as a wall and forming a pattern to maintain an adequate lap. Walls are bonded to provide maximum distribution of the load evenly throughout the wall. In order to achieve the bonding requirements at some point blocks were cut into different sizes. At the site the block wall had the thickness of 230mm.

There are several types of bonding applicable in construction activities, but the type of bonding adopted at the site was stretcher bonding for wall construction. Stretcher bond involved arranging blocks exposing their long narrow sides in stretcher courses whereby one block course overlap the other by half of its length.

Purpose of bonding

- Obtain maximum strength while distributing the load to the walls.
- Ensure lateral stability and resistance to side thrust.
- To create an acceptable appearance.



Plate 3.21: Showing bonding of blocks in a block work

3.1.11 Plastering

Plastering refers to the process of applying a thin layer of mortar on the walls or ceiling surfaces in order to obtain an even smooth, regular and clean surface. Plastering contains a mixture of cement, sand and a required amount of water.

Objectives of Plastering

- To protect the external surface against penetration of rain water and other atmospheric agencies.
- To provide a smooth surface on which dust and dirty cannot lodge.
- To provide a decorative effect.
- To control inferior materials or defective workmanship.

Procedures followed during plastering

- The surface to be plastered was cleaned by removing the protruded mortar during the brickwork through hammering and hacking. After cleaning, the plaster dabs of about 15mm thick were fixed on the wall surface in order to obtain a uniform surface.
- A plumb bob was used to check the uniformity of the dabs thickness. After obtaining verticality the dabs were left to dry, ready for the plastering. This helps to obtain a uniform thickness of the plaster at the end.
- The required amount of sand to be used in plastering was sieved by a 5mm sieve to obtain a uniform fine aggregate that is to be used in the mixture of plaster.

- The mortar to be used for plaster was mixed in a required amount of sand cement ratio with an appropriate amount of water to obtain the required mortar ratio. The mixing of plaster was done manually.
- The walls to be plastered were watered so as to remove dust and prevent drying of the plaster before proper application. This is also to ensure there is a strong bond between the plaster and the wall surface.
- The first coat of about 10mm thick plaster was applied by means of a steel trowel and then leveled by a wooden float and a steel straightening edge. The second coat of about 5mm thick plaster was applied on top of the first coat and then leveled by a wooden float followed by a steel float to make a 15mm thick plaster which matched with the pre-fixed dabs.
- After the application of plaster, it was then left to dry and from the next day it was daily watered for 3 consecutive weeks, so as to make it stronger.



Plate 3.22: Showing a plastered wall

- The plastered wall should be very well cured in order to achieve maximum bonding on the plaster and to avoid plaster defects.

Plastering defects

- Peeling effect. Some portion of the plaster wears off due to the failure to achieve a successive bond between the plaster coat and the wall, also due to poor mixing of the plaster.
- Cracking effect. Thermal movement may cause cracking on the plastered wall. This is due to low quality of the sand used in the preparation of plaster.

CHAPTER FOUR

4.1 CONCLUSION

The industrial training period was conducted for eight weeks at The 8 Construction Limited site at Temeke and ended successfully thanks to the good support from the staff members, site engineers and all other constructors, which enabled us to gain new skills due to the availability of various learning resources.

The training period has impacted a lot of practical knowledge, experience and exposure into the practical application of our field of study in real life. We also gained knowledge on how to read and analyze different structural and engineering drawings, and practical use of different construction equipment. We even got a chance of observing some of the stages followed during the construction of a building and thus enhance our theoretical knowledge.

4.2 CHALLENGES

- Delay of the construction process due to the delay in arrival of the construction materials on site, example plumbing materials, aggregates and sand.
- Insufficient working equipment such as helmets to the workers and hence reduce the safety environment to the workers.
- Inadequate skilled experts i.e. engineers, architects and quantity surveyors which led to the slow progress of the project.
- Existence of a large number of students from various universities accommodated by the particular company which led to improper management and supervision of the students by the site engineers.

4.3 RECOMMENDATIONS

- The materials to be used in the construction process such as plumbing materials, aggregates and sand should be readily available within the site so as to avoid unnecessary delay caused by their absence.
- There should be an increase of safety equipment so as to ensure maximum health safety at the site during construction activities.
- The tools and equipment used in the construction process should be increased so as to increase the working progress.
- The industrial training period should be increased in order to attain more knowledge concerning construction activities.

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